

LOGISTICS DELIVERY PERFORMANCE ANALYSIS

SQL-Based Exploratory & Performance Analysis
25,000 Logistics Delivery Records | PostgreSQL | Data Analytics

Project Title

Logistics Delivery Performance Analysis

Business Problem Statement

The objective of this project is to analyze 25,000 logistics delivery records to evaluate operational performance, identify factors contributing to delivery delays and failed deliveries, compare delivery partners, assess the impact of weather conditions and distance on delivery performance, analyze delivery costs, and understand customer satisfaction through delivery ratings.

This analysis helps logistics companies:

- Improve on-time delivery performance
- Reduce delivery delays and failures
- Optimize delivery costs
- Identify top-performing delivery partners
- Increase customer satisfaction
- Make data-driven business decisions

Dataset Overview

Total Records	25,000
Total Columns	15
Primary Key	delivery_id
Dataset Format	CSV

Database	PostgreSQL
Domain	Logistics and Supply Chain
Analysis Type	Exploratory and Performance Analysis

Columns Included

- delivery_id
- delivery_partner
- package_type
- vehicle_type
- delivery_mode
- region
- weather_condition
- distance_km
- package_weight_kg
- delivery_time_hours
- expected_time_hours
- delayed
- delivery_status
- delivery_rating
- delivery_cost

Tools Used

- PostgreSQL
- pgAdmin 4
- Microsoft Excel / CSV
- Microsoft Word
- GitHub
- LinkedIn

SQL Skills Used

- SELECT
- WHERE

- DISTINCT
 - COUNT()
 - COUNT(DISTINCT)
 - AVG()
 - SUM()
 - MIN()
 - MAX()
 - ROUND()
 - CASE WHEN
 - GROUP BY
 - ORDER BY
 - LIMIT
 - Column Aliases (AS)
 - Conditional Aggregation
 - Percentage Calculations
 - Window Functions (RANK())
 - Data Categorization
 - Business KPI Calculations
-

KPIs Analyzed

- Total Deliveries
 - Delay Percentage
 - Failure Rate
 - Delivery Status Distribution
 - Average Delivery Time
 - Average Delivery Cost
 - Average Delivery Rating
 - Cost per Kilometer
 - Cost per Kilogram
 - Delay Percentage by Weather Condition
 - Delay Percentage by Distance Category
-

SQL Analysis Screenshots

Basic SQL Queries (Q1–Q10)

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Servers

Delivery Logistics/postgres@PostgreSQL 18

No limit

Query Query History

```
19
20 -- ● Beginner Level (10 Questions)
21 -- Q1 Count total deliveries.
22 SELECT
23     COUNT(*)
24 FROM
25     DELIVERS;
26
27 -- Q2 Count unique delivery partners.
28 SELECT
29     COUNT(DISTINCT DELIVERY_PARTNER)
30 FROM
31     DELIVERS;
32
33 -- Q3 Show all distinct regions.
34 SELECT DISTINCT
35     REGION
36 FROM
37     DELIVERS;
38
39 -- Q4 Average delivery rating.
40 SELECT
41     AVG(DELIVERY_RATING)
42 FROM
43     DELIVERS;
44
45 -- Q5 Maximum delivery cost.
46 SELECT
47     MAX(DELIVERY_COST) AS MAX_COST
48 FROM
49     DELIVERS;
50
```

Total rows: 2935 Query complete 00:00:00.162

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No limit

Query Query History

```
51 -- Q6 Minimum delivery time.
52 SELECT
53     MIN(DELIVERY_TIME_HOURS)
54 FROM
55     DELIVERS;
56
57 -- Q7 Count delayed deliveries.
58 SELECT
59     COUNT(DELAYED) AS DELAYED_YES
60 FROM
61     DELIVERS
62 WHERE
63     DELAYED = 'Yes';
64
65 -- Q8 Count failed deliveries.
66 SELECT
67     COUNT(*) AS FAILED_DELIVERYS
68 FROM
69     DELIVERS
70 WHERE
71     DELIVERY_STATUS = 'Failed';
72
73 -- Q9 Average package weight.
74 SELECT
75     AVG(PACKAGE_WEIGHT_KG)
76 FROM
77     DELIVERS;
78
79 -- Q10 Average distance.
80 SELECT
81     AVG(DISTANCE_KM)
82
```

Total rows: 2935 Query complete 00:00:00.162

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No limit

Query Query History

```
79 -- Q10 Average distance.
80 SELECT
81     AVG(DISTANCE_KM)
82 FROM
83     DELIVERS;
84
```

Intermediate SQL Queries (Q11–Q25)

```
84
85 -- 🟡 Intermediate Level (15 Questions)
86 -- Q11 Delay percentage.
87 SELECT
88     ROUND(
89         COUNT(
90             CASE
91                 WHEN DELAYED = 'Yes' THEN 1
92             END
93         ) * 100.0 / COUNT(DELAYED),
94         2
95     ) AS DELAY_RATE
96 FROM
97     DELIVERS
98 -- Q12 Average cost by region.
99 SELECT
100     AVG(DELIVERY_COST),
101     REGION
102 FROM
103     DELIVERS
104 GROUP BY
105     REGION;
106
107 -- Q13 Average rating by delivery partner.
108 SELECT
109     ROUND(AVG(DELIVERY_RATING), 2) AS AVG_RATING,
```

Total rows: 2935 Query complete 00:00:00.162

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Query Query History

```
107 -- Q13 Average rating by delivery partner.
108 SELECT
109     ROUND(AVG(DELIVERY_RATING), 2) AS AVG_RATING,
110     DELIVERY_PARTNER
111 FROM
112     DELIVERS
113 GROUP BY
114     DELIVERY_PARTNER;
115
116 -- Q14 Top 5 most expensive deliveries.
117 SELECT
118     DELIVERY_COST AS TOP_5_EXPENSIVE
119 FROM
120     DELIVERS
121 ORDER BY
122     DELIVERY_COST DESC
123 LIMIT
124     5;
125
126 -- Q15 Total deliveries by package type.
127 SELECT
128     COUNT(DELIVERY_ID),
129     PACKAGE_TYPE
130 FROM
131     DELIVERS
132 GROUP BY
133     PACKAGE_TYPE;
134
135 -- Q16 Average delivery time by vehicle type.
136 SELECT
137     AVG(DELIVERY_TIME_HOURS),
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
135 -- Q16 Average delivery time by vehicle type.
136 SELECT
137     AVG(DELIVERY_TIME_HOURS),
138     VEHICLE_TYPE
139 FROM
140     DELIVERS
141 GROUP BY
142     VEHICLE_TYPE;
143
144 -- Q17 Delayed deliveries by weather condition.
145 SELECT
146     WEATHER_CONDITION,
147     COUNT(DELAYED)
148 FROM
149     DELIVERS
150 WHERE
151     DELAYED = 'Yes'
152 GROUP BY
153     WEATHER_CONDITION
154 -- Q18 Failure rate by delivery partner.
155 SELECT
156     ROUND(
157         COUNT(
158             CASE
159                 WHEN DELIVERY_STATUS = 'Failed' THEN 1
160             END
161         ) * 100.0 / COUNT(DELIVERY_STATUS),
162         2
163     ) AS FAILURE_RATE,
164     DELIVERY_PARTNER
165 FROM
166     DELIVERS
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
154 -- Q18 Failure rate by delivery partner.
155 SELECT
156     ROUND(
157         COUNT(
158             CASE
159                 WHEN DELIVERY_STATUS = 'Failed' THEN 1
160             END
161         ) * 100.0 / COUNT(DELIVERY_STATUS),
162         2
163     ) AS FAILURE_RATE,
164     DELIVERY_PARTNER
165 FROM
166     DELIVERS
167 GROUP BY
168     DELIVERY_PARTNER;
169 -- Q19 Average cost by delivery mode.
170 SELECT
171     AVG(DELIVERY_COST),
172     DELIVERY_MODE
173 FROM
174     DELIVERS
175 GROUP BY
176     DELIVERY_MODE;
177 -- Q20 Top 3 partners with highest ratings.
178 SELECT DISTINCT
179     DELIVERY_PARTNER,
180     DELIVERY_RATING
181 FROM
182     DELIVERS
183 ORDER BY
184     DELIVERY_RATING DESC
185 LIMIT:
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
188 -- Q21 Most used vehicle type.
189 SELECT
190     COUNT(VEHICLE_TYPE),
191     VEHICLE_TYPE
192 FROM
193     DELIVERS
194 GROUP BY
195     VEHICLE_TYPE
196 ORDER BY
197     COUNT DESC
198 LIMIT
199     1;
200
201 -- Q22 Region with highest delay rate.
202 SELECT
203     REGION,
204     ROUND(
205         COUNT(
206             CASE
207                 WHEN DELAYED = 'Yes' THEN 1
208             END
209         ) * 100.0 / COUNT(DELAYED),
210         2
211     ) AS DELAY_RATE
212 FROM
213     DELIVERS
214 GROUP BY
215     REGION
216 ORDER BY
217     DELAY_RATE DESC;
218
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
219 -- Q23 Weather with lowest average rating.
220 SELECT DISTINCT
221     WEATHER_CONDITION,
222     ROUND(AVG(DELIVERY_RATING), 2) AS AVG_RATING
223 FROM
224     DELIVERS
225 GROUP BY
226     WEATHER_CONDITION
227 ORDER BY
228     AVG_RATING ASC;
229
230 -- Q24 Average distance by package type.
231 SELECT
232     AVG(DISTANCE_KM),
233     PACKAGE_TYPE
234 FROM
235     DELIVERS
236 GROUP BY
237     PACKAGE_TYPE;
238
239 -- Q25 Average rating by weather.
240 SELECT
241     AVG(DELIVERY_RATING),
242     WEATHER_CONDITION
243 FROM
244     DELIVERS
245 GROUP BY
246     WEATHER_CONDITION;
247
248 -- ---
249 -- ● Advanced Level (15 Questions)
```

Total rows: 2935 Query complete 00:00:00.162

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Advanced SQL Queries (Q26–Q38)

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
249 -- ● Advanced Level (15 Questions)
250 -- Q26 Cost per kilometer.
251 SELECT
252     DISTANCE_KM,
253     DELIVERY_COST,
254     ROUND(DELIVERY_COST / DISTANCE_KM, 2) AS PER_KM
255 FROM
256     DELIVERS
257 -- Q27 Compare actual vs expected time.
258 SELECT
259     DELIVERY_TIME_HOURS,
260     EXPECTED_TIME_HOURS
261 FROM
262     DELIVERS
263 -- Q28 Top 5 costly delayed deliveries.
264 SELECT
265     DELAYED,
266     DELIVERY_COST AS TOP_5_DELAY
267 FROM
268     DELIVERS
269 WHERE
270     DELAYED = 'Yes'
271 ORDER BY
272     TOP_5_DELAY DESC
273 LIMIT
274     5;
275
276 -- Q29 Delivery status distribution %.
277 SELECT
278     ROUND(
279         COUNT(
280             CASE
281                 WHEN DELIVERY_STATUS = 'Failed' THEN 1
282             END
283         ) * 100.0 / COUNT(DELIVERY_STATUS),
284         2
285     ) AS FAILED,
286     ROUND(
287         COUNT(
288             CASE
289                 WHEN DELIVERY_STATUS = 'Delivered' THEN 1
290             END
291         ) * 100.0 / COUNT(DELIVERY_STATUS),
292         2
293     ) AS DELIVERED
294 FROM
295     DELIVERS;
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
276 -- Q29 Delivery status distribution %.
277 SELECT
278     ROUND(
279         COUNT(
280             CASE
281                 WHEN DELIVERY_STATUS = 'Failed' THEN 1
282             END
283         ) * 100.0 / COUNT(DELIVERY_STATUS),
284         2
285     ) AS FAILED,
286     ROUND(
287         COUNT(
288             CASE
289                 WHEN DELIVERY_STATUS = 'Delivered' THEN 1
290             END
291         ) * 100.0 / COUNT(DELIVERY_STATUS),
292         2
293     ) AS DELIVERED
294 FROM
295     DELIVERS;
296
297 -- Q30 Partner with highest failed deliveries.
298 SELECT
299     DELIVERY_PARTNER,
300     COUNT(DELIVERY_STATUS) FAIL_DELIVERS
301 FROM
302     DELIVERS
303 WHERE
304     DELIVERY_STATUS = 'Failed'
305 GROUP BY
306     DELIVERY_PARTNER
307 ORDER BY
308     FAIL_DELIVERS DESC;
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
303 WHERE
304     DELIVERY_STATUS = 'Failed'
305 GROUP BY
306     DELIVERY_PARTNER
307 ORDER BY
308     FAIL_DELIVERS DESC
309 LIMIT
310     1;
311
312 -- Q31 Average rating of delayed vs non-delayed.
313 SELECT
314     DELIVERY_STATUS,
315     ROUND(AVG(DELIVERY_RATING), 2)
316 FROM
317     DELIVERS
318 GROUP BY
319     DELIVERY_STATUS;
320
321 -- Q32 Most profitable delivery mode (highest avg cost).
322 SELECT
323     DELIVERY_MODE,
324     ROUND(AVG(DELIVERY_COST), 2) AS AVG_COST
325 FROM
326     DELIVERS
327 GROUP BY
328     DELIVERY_MODE
329 ORDER BY
330     AVG_COST DESC
331 LIMIT
332     1;
333
334 -- Q33 Weather impact on delay percentage.
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
334 -- Q33 Weather impact on delay percentage.
335 SELECT
336     WEATHER_CONDITION,
337     ROUND(
338         COUNT(
339             CASE
340                 WHEN DELAYED = 'Yes' THEN 1
341             END
342         ) * 100.0 / COUNT(DELAYED),
343         2
344     )
345 FROM
346     DELIVERS
347 GROUP BY
348     WEATHER_CONDITION;
349
350 -- Q34 Package type with highest cost per kg.
351 SELECT DISTINCT
352     PACKAGE_TYPE,
353     ROUND(AVG(DELIVERY_COST / PACKAGE_WEIGHT_KG), 2) AS COST_PER_KG
354 FROM
355     DELIVERS
356 GROUP BY
357     PACKAGE_TYPE
358 ORDER BY
359     COST_PER_KG DESC
360 LIMIT
361     1;
362
363 -- Q35 Vehicle type with fastest average delivery.
364 SELECT
365     VEHICLE_TYPE
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
-- Q35 Vehicle type with fastest average delivery.
SELECT
    VEHICLE_TYPE,
    ROUND(AVG(DELIVERY_TIME_HOUR), 2) AS AVG_TIME
FROM
    DELIVERS
GROUP BY
    VEHICLE_TYPE
ORDER BY
    AVG_TIME ASC
LIMIT
    1;

-- Q36 Rank partners by rating using RANK().
SELECT
    DELIVERY_PARTNER,
    ROUND(AVG(DELIVERY_RATING), 2) AS RATING,
    RANK() OVER (
        ORDER BY
            (DELIVERY_PARTNER) DESC
    )
FROM
    DELIVERS
GROUP BY
    DELIVERY_PARTNER
-- Q37 Conditional CASE categorization for distance:
-- Short (<50 km)
-- Medium (50-150 km)
-- Long (>150 km)
SELECT
    DISTANCE_KM,
```

Total rows: 2935 Query complete 00:00:00.162

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Delivery Logistics/postgres@PostgreSQL 18

Query Query History

```
-- Medium (50-150 km)
-- Long (>150 km)
SELECT
    DISTANCE_KM,
    CASE
        WHEN DISTANCE_KM < 50 THEN 'Short'
        WHEN DISTANCE_KM BETWEEN 50 AND 150 THEN 'Medium'
        ELSE 'Long'
    END
FROM
    DELIVERS;

-- Q38 Delay percentage by distance category.
SELECT
    CASE
        WHEN DISTANCE_KM < 50 THEN 'Short'
        WHEN DISTANCE_KM BETWEEN 50 AND 150 THEN 'Medium'
        ELSE 'Long'
    END AS DISTANCE_CATG,
    ROUND(
        COUNT(
            CASE
                WHEN DELAYED = 'Yes' THEN 1
            END
        ) * 100.0 / COUNT(DELAYED),
        2
    ) AS DELAY_PERCENT
FROM
    DELIVERS
GROUP BY
    DISTANCE_KM;
```

Total rows: 2935 Query complete 00:00:00.162

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Key Insights

1. A measurable percentage of deliveries experienced delays, highlighting opportunities to improve on-time performance.
2. Certain delivery partners had higher failure rates than others.
3. Delayed deliveries generally received lower customer ratings.
4. Weather conditions significantly impacted delivery delays.
5. Delivery costs varied across regions, package types, and delivery modes.
6. Some vehicle types consistently delivered faster than others.
7. Longer-distance deliveries showed a higher likelihood of delays.
8. Specific package types generated higher costs per kilogram.
9. Delivery mode influenced both cost and delivery speed.
10. Operational performance varied across partners and regions.

Conclusion

This project demonstrates how SQL can be used to transform raw logistics data into meaningful business insights. By analyzing delivery delays, failures, costs, and customer ratings, the project identified key factors affecting logistics performance, including weather conditions, delivery distance, vehicle type, and delivery partner performance.

The findings can help logistics companies optimize operations, reduce delays, improve customer satisfaction, and make informed, data-driven decisions. This project also showcases strong SQL and analytical skills relevant to entry-level Data Analyst roles.

Project Files Included

- logistics_delivery_analysis.sql
 - logistics_delivery_data.csv
 - Logistics_Delivery_Analysis_Report.pdf
 - README.md
 - screenshots/
-

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